



DIGITAL LOGIC AND CIRCUITS

Logic Gates

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Digital Electronics involves circuits and systems in which there are only two possible States. These states are represented by two different voltage levels: A High and a Low Voltage Level.

Positive Logic:

High = 1

Low = 0

(mostly used)

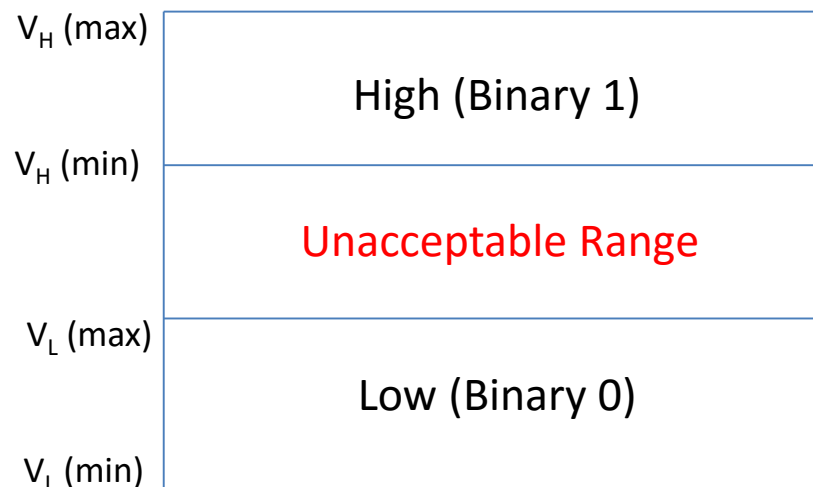
Negative Logic:

High = 0

Low = 1

Logic Levels:

The voltages used to represent a '1' or '0' are called logic levels.





Logic Circuits:

The manner in which a digital circuit responds to an input is referred to as a circuit logic.

In most basic logic circuits, logic gates are the fundamental building blocks from which all other Logic circuits and digital systems are constructed.

The digital circuit ICs are constructed using the following logic gates:

1) Basic Gates:

- NOT Gate or Inverter
- AND Gate
- OR Gate

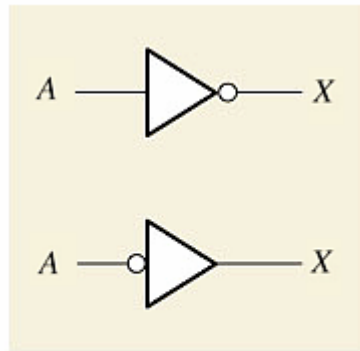
2) Universal Gates:

- NAND Gate
- NOR Gate

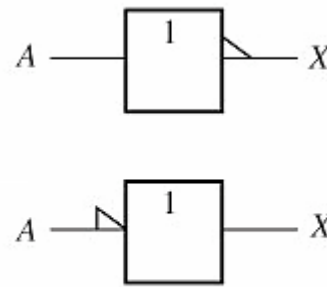
3) Exclusive Gates:

- Exclusive-OR Gate
- Exclusive-NOR Gate

The Inverter

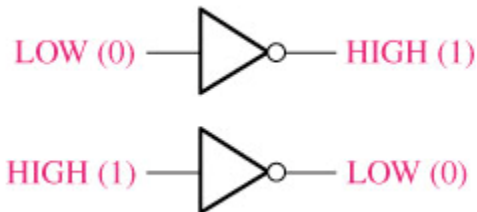


Distinctive shape symbols



Rectangular outline symbols

The output of an inverter is always the complement (opposite) of the input.



When the input is LOW, the output is HIGH
When the input is HIGH, the output is LOW

A	X
0	1
1	0

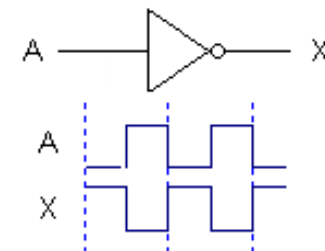
$$X = \bar{A}$$

Truth table

Boolean expression

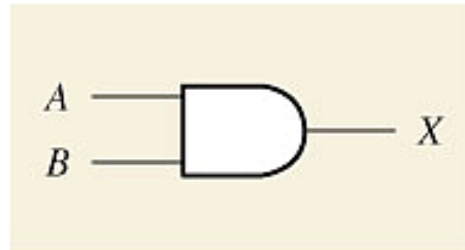
0 = LOW

1 = HIGH



Pulsed waveforms

The AND Gate

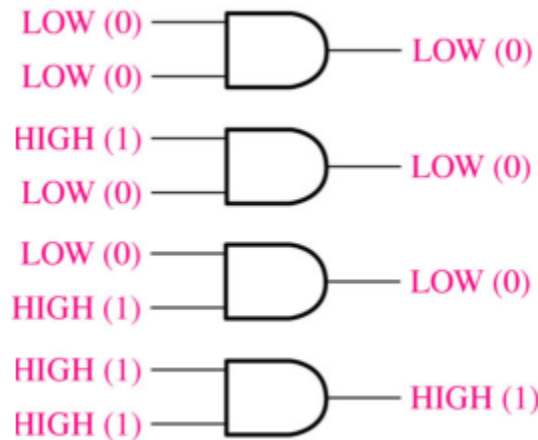


Distinctive shape symbol



Rectangular outline symbol

The output of an AND gate is HIGH only when all inputs are HIGH.



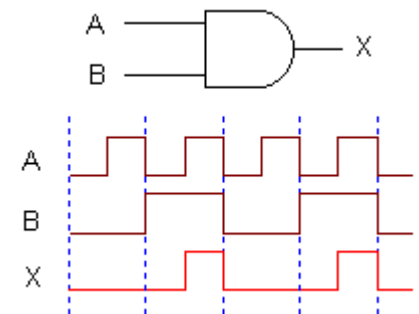
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

Truth table

0 = LOW
1 = HIGH

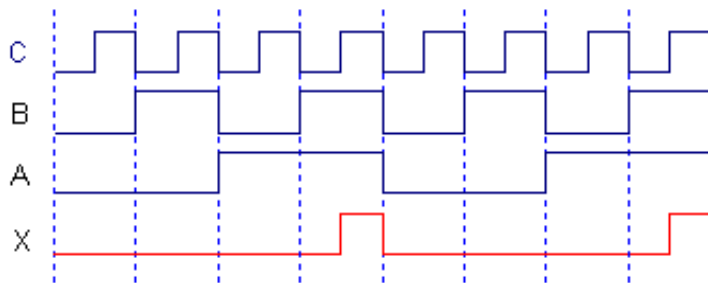
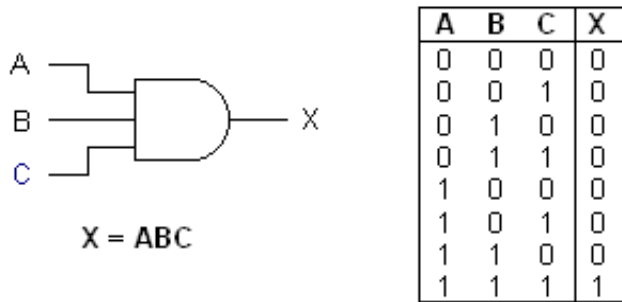
$$X = AB$$

Boolean expression

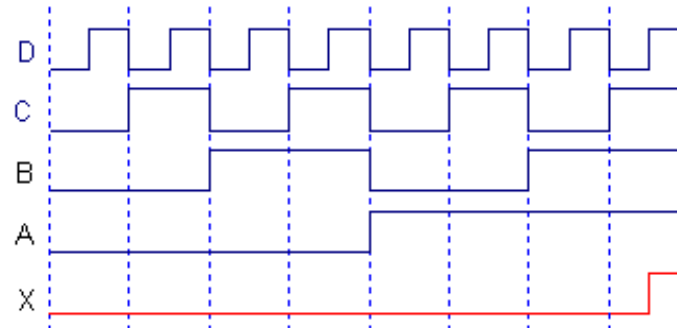
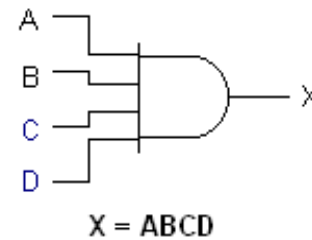


Pulsed waveforms

The AND Gate

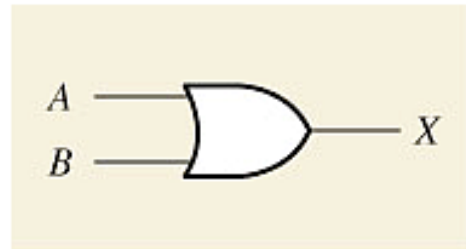


3-Input AND Gate

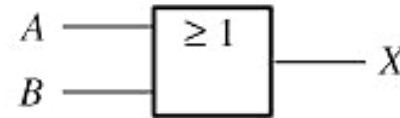


4-Input AND Gate

The OR Gate

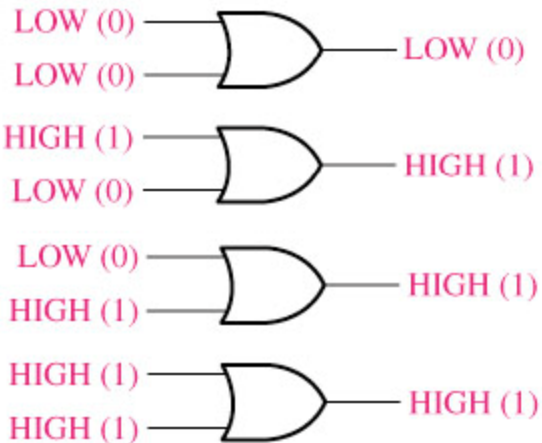


Distinctive shape symbol



Rectangular outline symbol

The output of an OR gate is HIGH whenever one or more inputs are HIGH



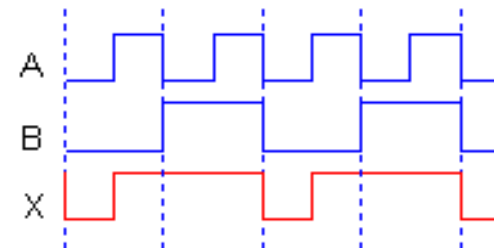
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Truth
table

0 = LOW
1 = HIGH

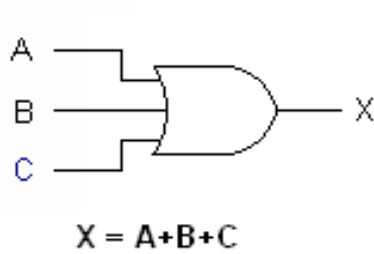
$$X = A + B$$

Boolean expression

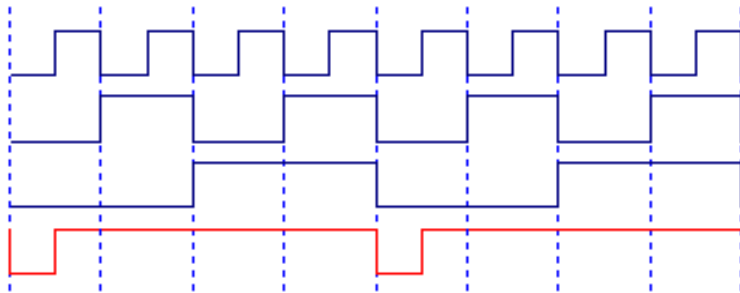


Pulsed waveforms

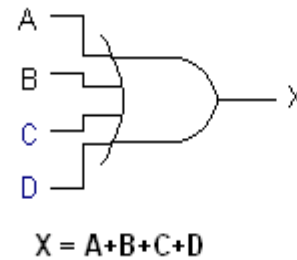
The OR Gate



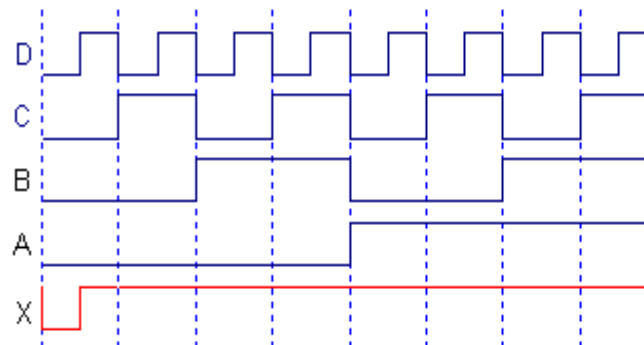
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



3-Input OR Gate

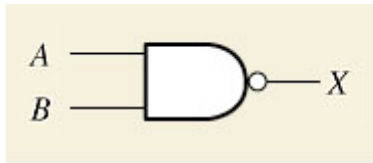


A	B	C	D	X
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

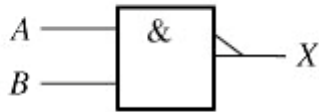


4-Input OR Gate

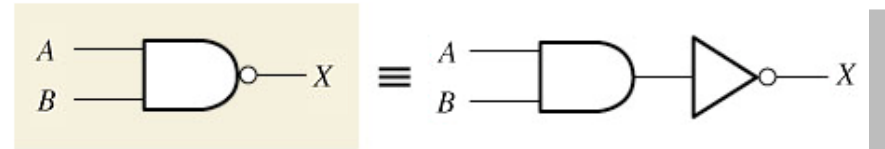
The NAND Gate



Distinctive shape symbol



Rectangular outline symbol



The output of a NAND gate is HIGH whenever one or more inputs are LOW.

A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

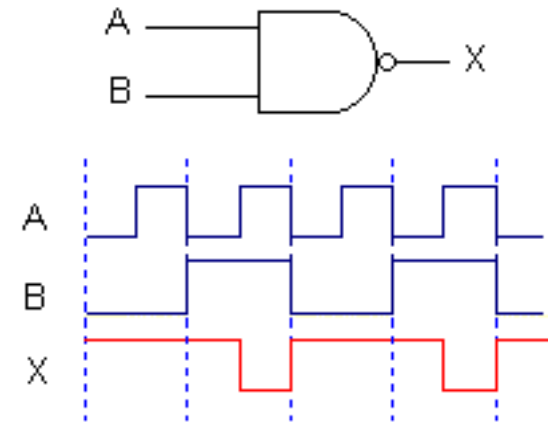
Truth table

0 = LOW

1 = HIGH

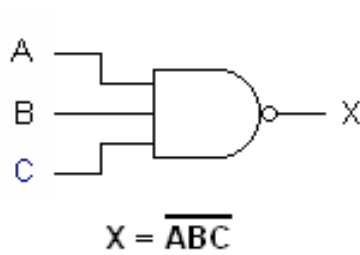
$$X = \overline{AB}$$

Boolean expression

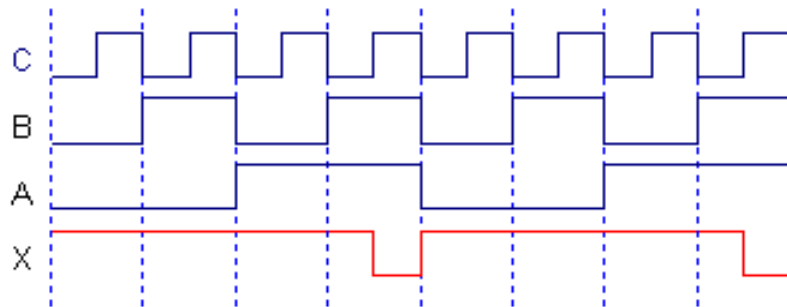


Pulsed waveforms

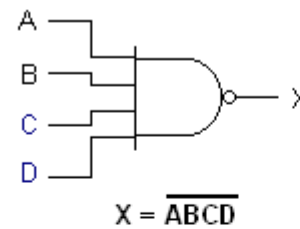
The NAND Gate



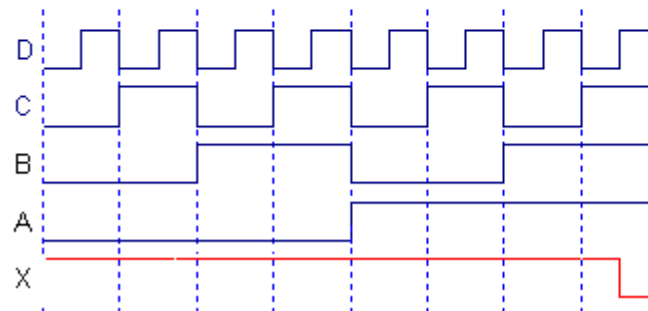
A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



3-Input NAND Gate

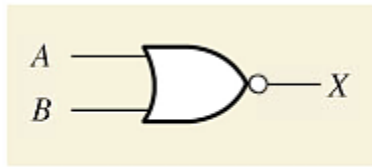


A	B	C	D	X
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
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1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

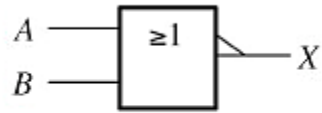


4-Input NAND Gate

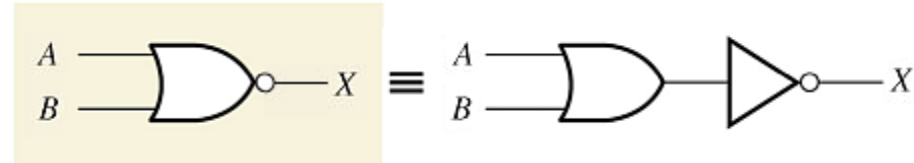
The NOR Gate



Distinctive shape symbol



Rectangular outline symbol



The output of a NOR gate is LOW whenever one or more inputs are HIGH.

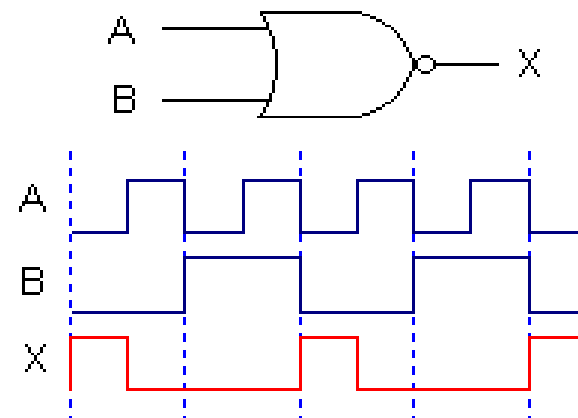
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

Truth
table

0 = LOW
1 = HIGH

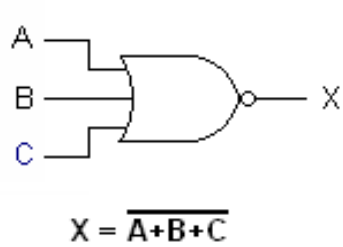
$$X = \overline{A + B}$$

Boolean
expression

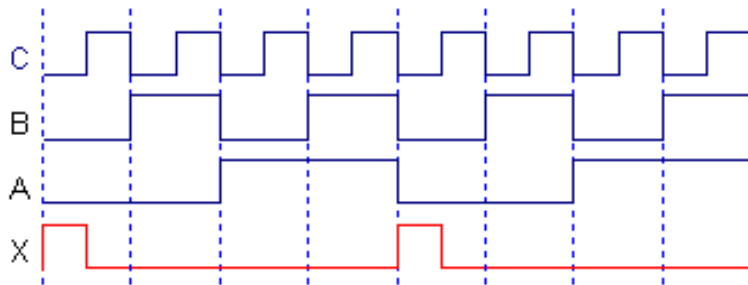


Pulsed waveforms

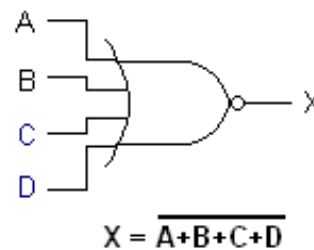
The NOR Gate



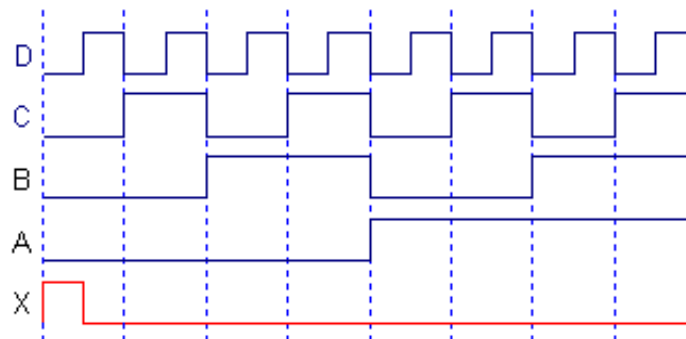
A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0



3-Input NOR Gate

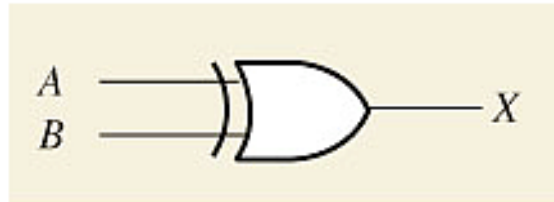


A	B	C	D	X
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

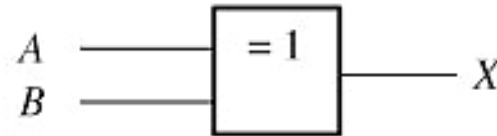


4-Input NOR Gate

Exclusive-OR and Exclusive-NOR Gates



Distinctive shape symbol



Rectangular outline symbol

The output of an XOR gate is HIGH whenever the two inputs are different.

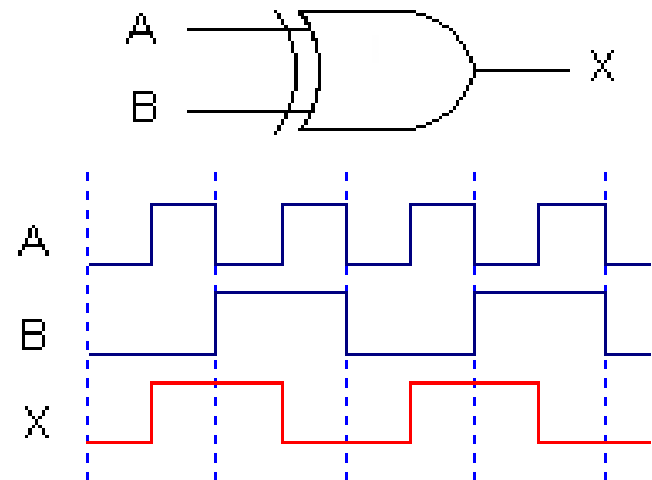
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

Truth
table

0 = LOW
1 = HIGH

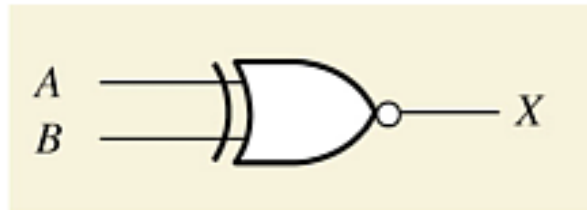
$$X = A \oplus B$$

Boolean
expression

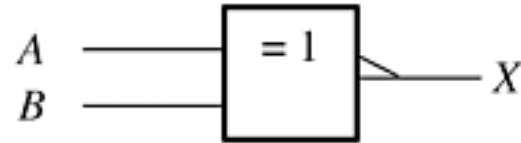


Pulsed waveforms

Exclusive-NOR Gate



Distinctive shape symbol



Rectangular outline symbol

The output of an XNOR gate is HIGH whenever the two inputs are identical.

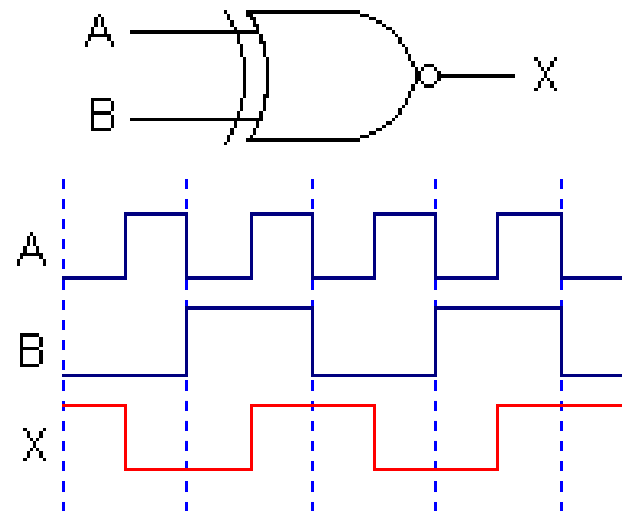
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

Truth
table

0 = LOW
1 = HIGH

$$X = \overline{A \oplus B}$$

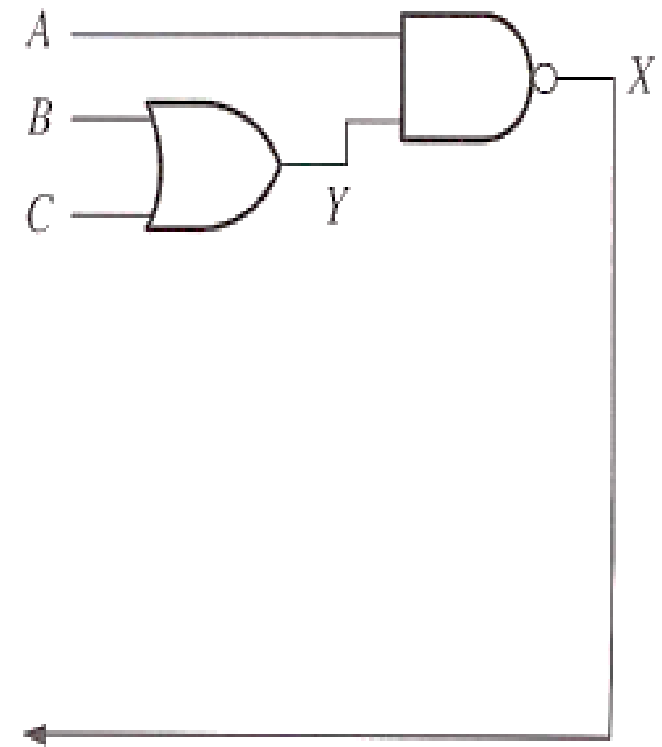
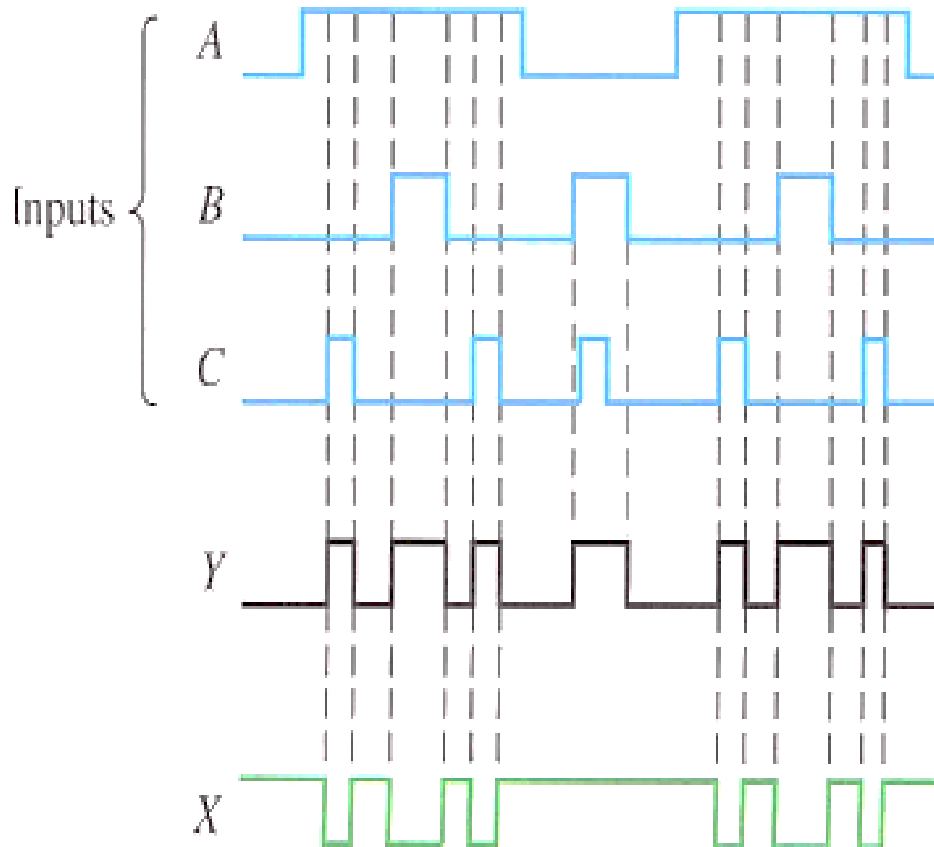
Boolean
expression



Pulsed waveforms

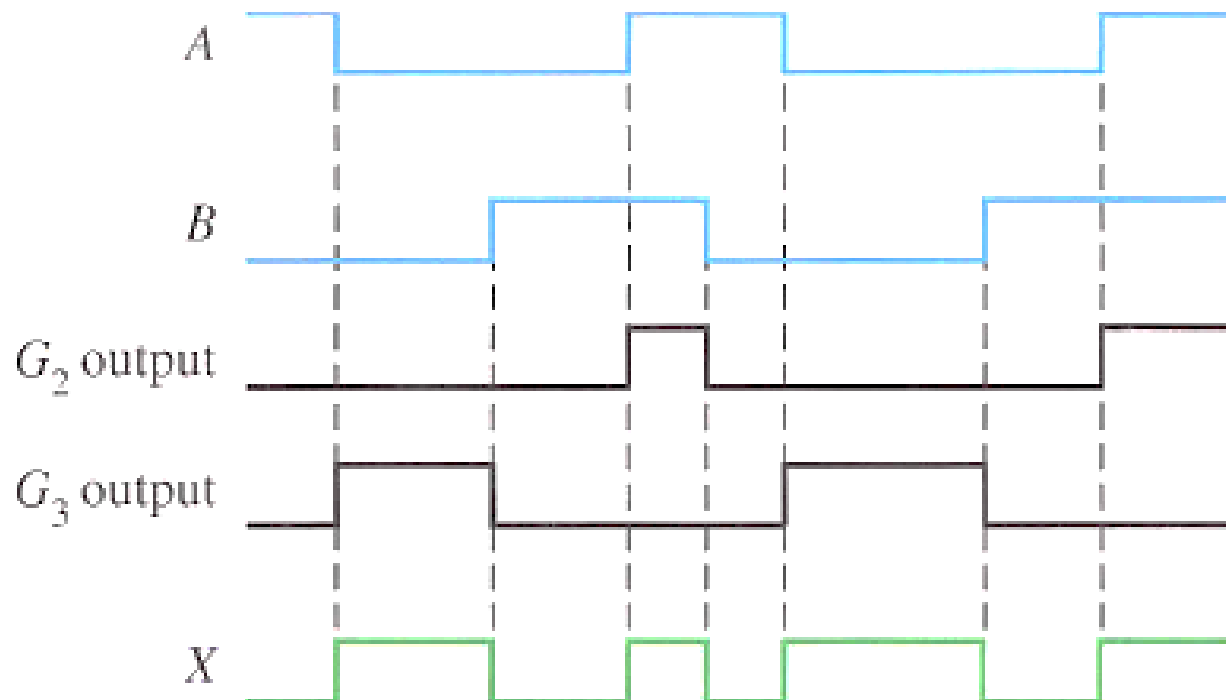
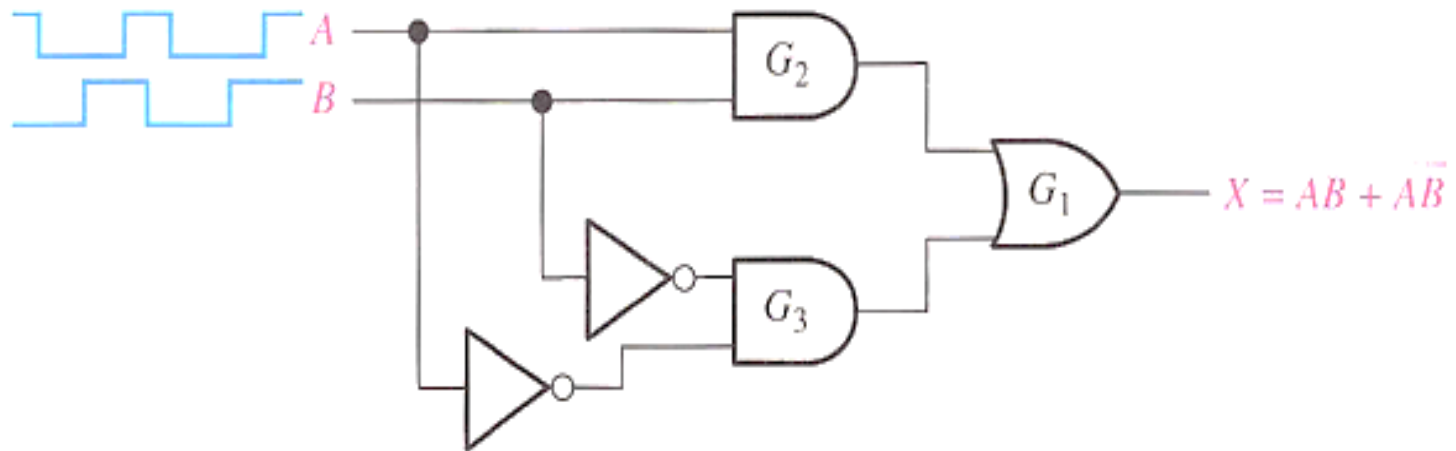
Logic Circuit Operation with Pulse Waveform Inputs:

Example 1:



$$X = \overline{A(B + C)} = \overline{AB} + \overline{AC}$$

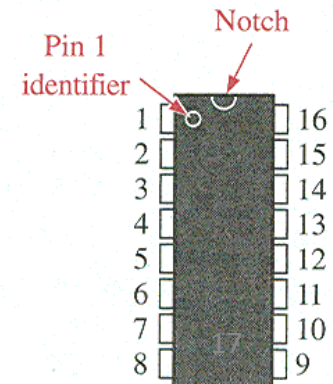
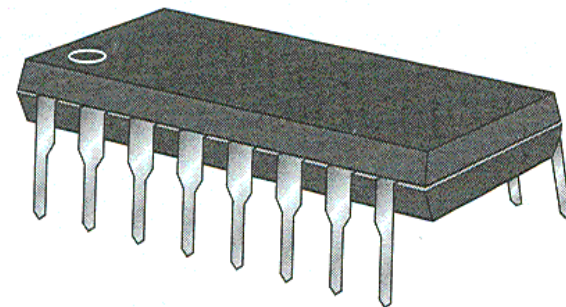
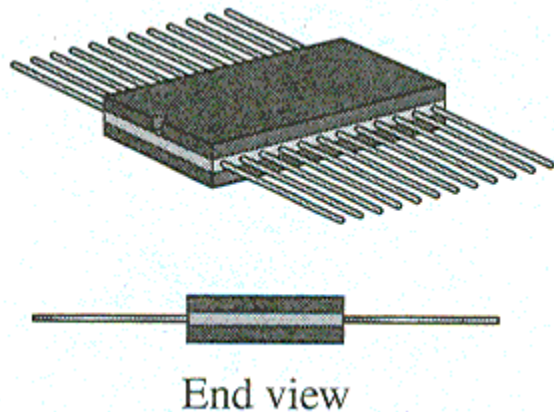
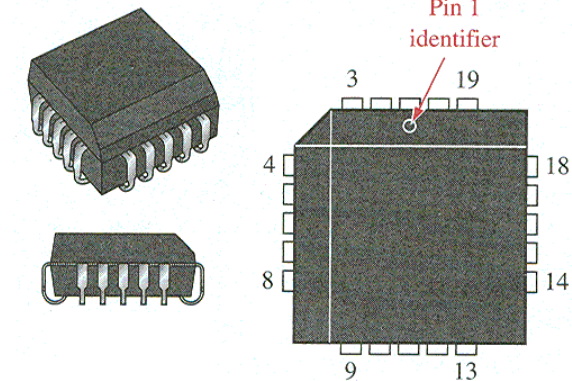
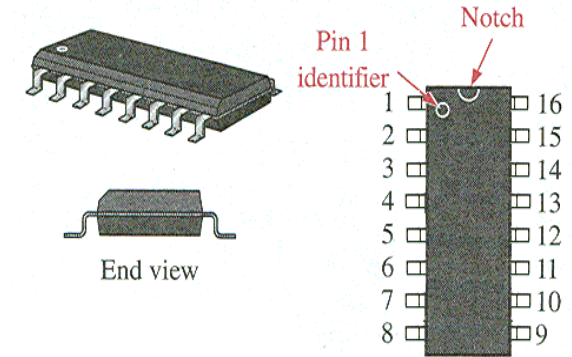
Example 2:



Fixed-Function Integrated Circuits

IC package styles

- Dual in-line package (DIP)
- Small-outline IC (SOIC)
- Flat pack (FP)
- Plastic-leaded chip carrier (PLCC)
- Leadless-ceramic chip carrier (LCCC)





Reference:

Mixed contents from books by Floyd; Mano.

Thanks